INCH-POUND
MIL-DTL-7793/16A
19 October 2012
SUPERSEDING
MIL-M-7793/16
15 December 1988

DETAIL SPECIFICATION SHEET

METER, TIME TOTALIZING, MINIATURE,
DIGITAL, 5 VDC, SQUARE

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

Reactivated after 19 October 2012 and may be
used for new and existing designs and acquisitions.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-7793.

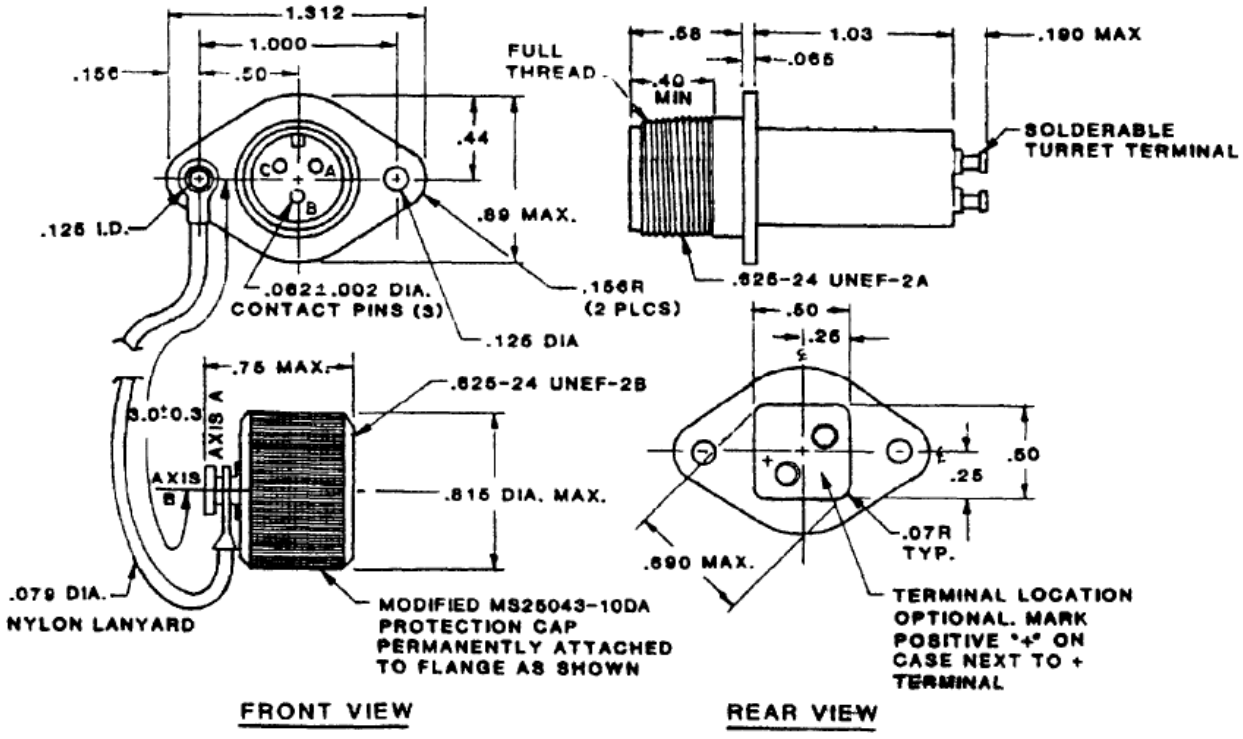


FIGURE 1. M7793/16-1.

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REQUIREMENTS:

1. Dimensions. Dimensions and configuration shall be in accordance with Figure 1. All dimensions are in inches. Unless otherwise specified, tolerances are ± .02 for two-place decimals and ±0.015 for three-place decimals.

2. Mounting flange. The mounting flange, threaded extension and threaded protection cap shall be constructed of aluminum alloy with an olive drab chromate coating over cadmium plating.

3. Protection cap. The protection cap shall be in accordance with MS25043-10DA per SAE-AS25043 except for the finish and the means of attachment. The finish shall be in accordance with Requirement 2 and the metallic chain shall be replaced with a nylon lanyard.

4. Contacts. The contacts shall meet all the requirements of silver plated solder contacts from SAE-AS50151.

PERFORMANCE REQUIREMENTS:

1. Dielectric strength. The meter shall be capable of withstanding the application of 600 VRMS (room) and 350 VRMS (altitude) between the terminals and the case.

2. Life accuracy. Life accuracy shall be ±0.1% from -65° C to +125° C and 4.5 to 10 VDC.

3. Power consumption. Power consumption at 5 VDC shall not exceed 2 milliwatts.

4. Ripple voltage. The meter shall operate normally when subjected to a 2 volt peak (4 volts peak-to-peak) ripple between 10 Hz and 10 KHz superimposed on 7.0 VDC.

5. Operating temperature range. The operating temperature range shall be from -65° C to +125° C.

6. Part number. The part number shall be M7793/16-1.

7. Time range. Total readout of the meters shall be 99999.99 hours.

8. Meter output. The output from the meter shall be in a serial binary coded decimal format. The output impedance shall be 100K ± 10%. A logical “zero” shall be between 0.0 and +0.2 volt. A logical “one” shall be between +3.3 and +6.6 volts. The output pin assignments are shown on figure 2 and the output data format is shown on figure 3.

9. Transient protection. No temporary or permanent degradation shall be produced in the meter when subjected to ±25 volt transients of 10 microsecond duration occurring at a 1 millisecond repetition rate.

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10. Durability of cap and threads. No mechanical or electrical degradation after 100 cycles in accordance with SAE-AS50151.

11. Operating voltage range. The operating voltage range shall be from 4.5 to 10 VDC.

12. Durability of connectors and contacts. No mechanical or electrical degradation after 100 cycles in accordance with SAE-AS50151. Operational accuracy shall be checked at the end of the test.

13. Weight. The weight shall be not greater than 1.0 ounce with the attached protection cap.

14. Tensile strength of lanyard. The lanyard shall withstand a tensile load of 25 pounds when applied to Axis “A” and Axis “B” as shown on figure 1.

15. Qualification tests. Qualification tests will be conducted in accordance with MIL-DTL-7793, except that the voltage, temperature and accuracy shall be as specified herein. Salt spray, moisture resistance, vibration, shock, electromagnetic compatibility and low temperature storage shall be conducted with the protective cap in place. Hermetic seal test is not applicable.

NOTES.

1. Elapsed time is read from the M7793/16-1 meter by mating the M7793/16-1 meter’s connector with the M7793/12-1 reader’s connector and operating the reader.
2. Digital readout and meter face plate per MIL-DTL-7793 are not applicable since the digital readout is provided by the separate M7793/12-1 reader.
3. The meter is designed to have an electrical ground and a continuous EMI/RFI shield with the threaded protection cap in place.

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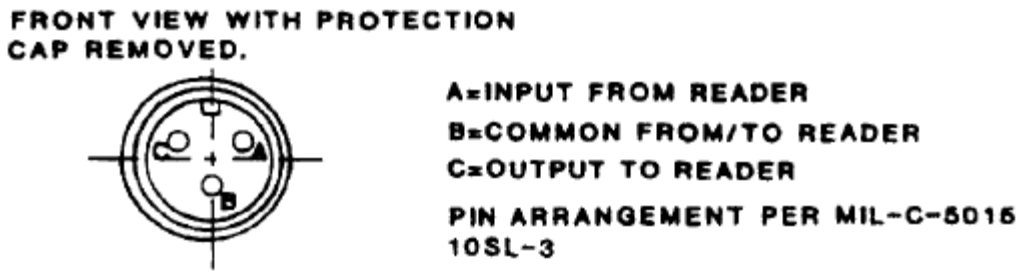


FIGURE 2. Readout pin assignment.

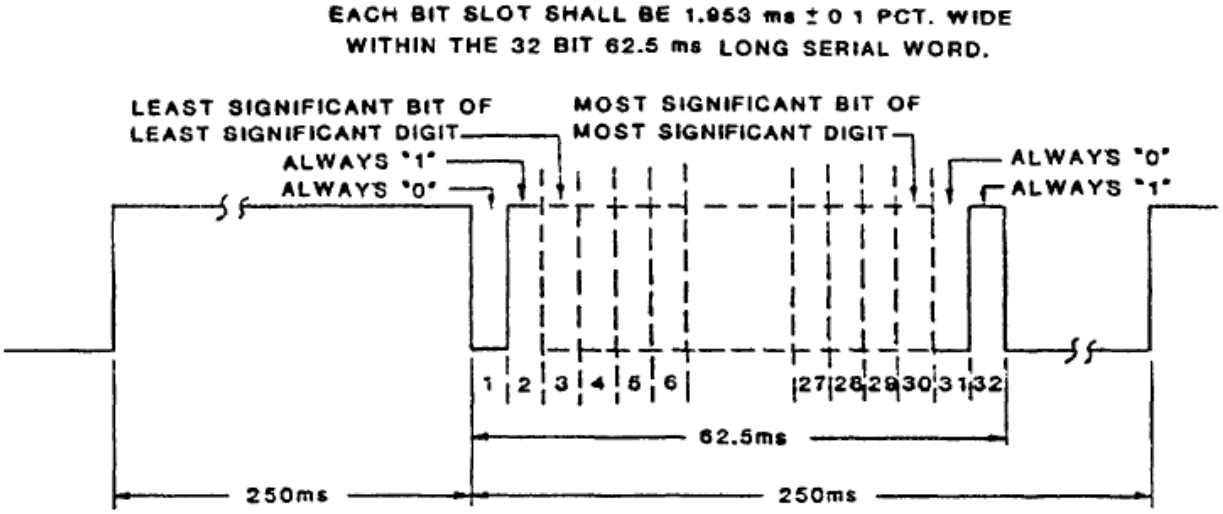


FIGURE 3. Output data format.

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Referenced documents. In addition to MIL-DTL-7793, this specification sheet references SAE-AS50151 and SAE-AS25043.

8. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - MI
Navy - AS
Air Force - 99

Preparing activity:
Navy - AS
(Project 6645-2012-018)

Review activities:
Navy – OS, SH
DLA – GS1

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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